

QIWEN(KYRA) JIAO

+1 6479161088 | ✉ kyraojiao@gmail.com
🌐 <https://www.linkedin.com/in/qiwen-jiao/> | 🌐 <https://jqiwen.com/>

EDUCATION

- Master of Engineering, Electrical and Computer Engineering** Sep. 2025 – Dec. 2026
University of Waterloo, Waterloo, ON, CA (GPA: 3.9/4.0)
- Bachelor of Applied Science, Honours Computer Science (Co-op)** Sep. 2020 – Jun. 2025
McMaster University, Hamilton, ON, CA (GPA: 3.7/4.0)

EXPERIENCES

- Software Development Engineer (Intern)** May 2026 - Aug 2026
Quantumera AI, Remote (U.S.)

- Built and deployed an AWS-based transportation data ingestion pipeline using SQS and EC2, integrating DynamoDB data and storing processed outputs in S3 with IAM-based access control.
- Created an ElastiCache Redis cluster to cache frequently accessed data, reducing end-to-end latency by 33%.
- Processed S3 data with EMR and Spark and indexed outputs in Elasticsearch to support downstream analytics and AI models.

- Software Development Engineer (Intern)** Aug. 2023 – Aug. 2024
Bosch (China) Investment Ltd, Suzhou, China

- Independently developed and deployed an enterprise-level web system integrating frontend and backend microservices on distributed servers, attracting 5k+ active users and handling 1M+ user requests in parallel.
- Designed and implemented reusable frontend components with Angular/Vue, improving UI consistency across 20+ modules. Utilized Pinia, RxJS, and Redux for state management and efficient data flow, reducing redundant logic by 30%.
- Developed backend services with Spring Boot and MySQL, implementing optimized queries and RESTful APIs documented via Swagger UI. Improved response time by 30% and reduced integration defects by 25%.
- Integrated CI/CD pipelines and Dockerized deployments using GitHub Actions under Linux environments, reducing release time by 40% and ensuring distributed reliability.
- Collaborated cross-functionally to design and test APIs, integrating basic NLP-based text parsing modules for data normalization and improving consistency across services; achieved a 99% on-time delivery rate following Agile practices.

- Automation Test Engineer (Intern)** Jun. 2021 – Dec. 2021
Bosch (China) Investment Ltd, Suzhou, China

- Developed and maintained automated test suites using JUnit, designing unit, integration, and regression testing to ensure system reliability and performance.
- Designed detailed test plans and scenarios based on product requirements, executing 200+ test cases, and identifying 50+ critical defects. Worked closely with developers to provide actionable feedback and improve overall product quality.
- Integrated automated tests into CI/CD pipelines with GitHub Actions, enabling continuous validation during builds. This reduced release cycle time by 30%, improved early defect detection.

PROJECTS

- Hammerly, Online Bidding Platform with AI customer support** Jan. 2026 – Aug. 2026
React, TypeScript, Zustand, Java, Spring Boot, Spring Security, PostgreSQL, Supabase, GCP, Microservices

- Led frontend development for a full-stack online bidding platform, building 5+ core user workflows including auction browsing, bidding, listing creation, profiles, and watchlists with React and SpringBoot.
- Designed a microservice architecture with 2 independently deployable Spring Boot services for core marketplace operations and AI customer support, implementing 15+ RESTful APIs with Spring Security, JWT authentication, and PostgreSQL for users, auctions, bids, and account data.
- Built and deployed a high-concurrency cloud AI platform with Docker, Kubernetes, and GCP, supporting 1,000+ concurrent users; conducted k6 load testing and monitored throughput, P95 latency, error rates, JVM metrics, and service health with Prometheus to identify and improve performance bottlenecks.

TECHNICAL SKILLS

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| Languages: | Python, Java, JavaScript, TypeScript, SQL |
| Backend & Frontend: | Spring Boot, Django, Node.js, React, Angular, Vue, REST APIs |
| AI & Data: | Generative AI, LLM APIs, RAG, NLP, Apache Spark, Elasticsearch, Redis |
| Cloud & DevOps: | AWS, GCP, Docker, Kubernetes, GitHub Actions, Linux |